

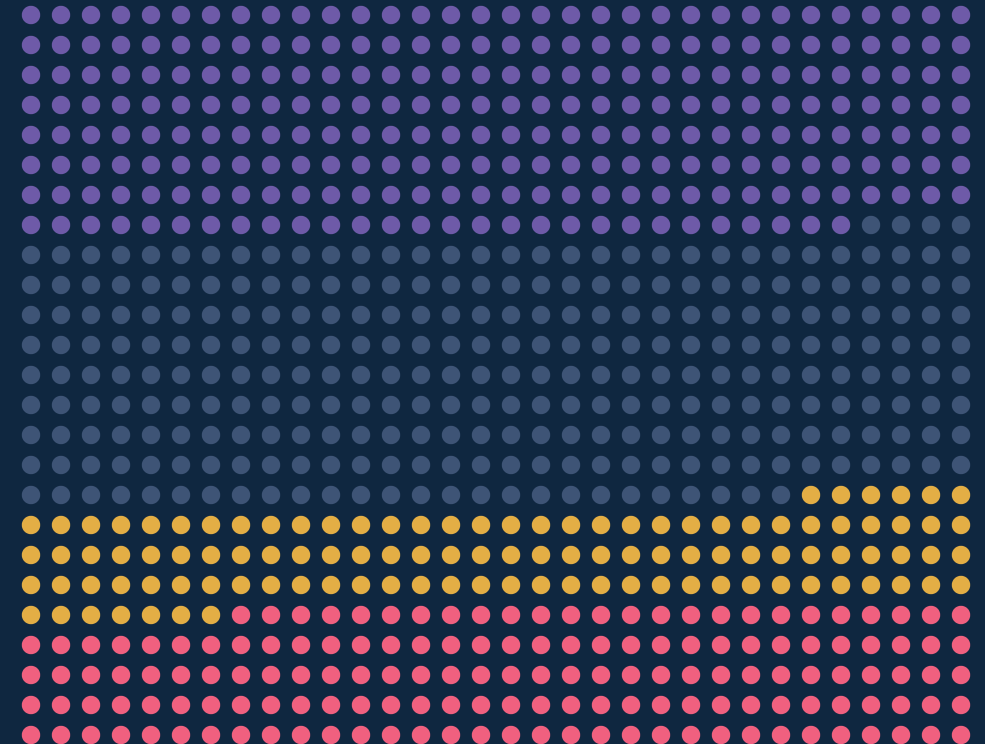
Lead qualification and conversion scoring

Finding the leads worth calling

How we took an unranked inbox of spam, tire-kickers and real buyers, and turned it into a scored, explainable queue that tells sales who to call next and why.

Product case study, September 2026

All figures come from a synthetic dataset of 800 leads built to rehearse the method.



● Spam 252 ● Nurture 286 ● Warm 109 ● Hot 153

Every dot is one lead from the last 12 months, sorted by the bucket our system assigns.

We turned an unranked lead inbox into a scored, explainable queue

The problem

Spam, vendor pitches and real buyers land in one queue. Sales answers them in arrival order, so a ready-to-buy team waits as long as a bot. A third of leads never get an outcome recorded, so nobody learns what a good lead looks like.

What we built

A two-stage system. A spam gate quarantines junk without deleting it. A conversion score ranks every genuine lead into Hot, Warm or Nurture, gives a calibrated probability, and explains itself in plain words. Every sales action flows back as training data.

What we found

Clean outcome logging mattered more than any model. A transparent points score captured most of the value. A learned model added a small lift in ranking and, more importantly, probabilities sales can trust.

0%

genuine leads wrongly quarantined, down from 34% in the first draft of the rules

92%

of resolved Hot leads reached a qualified opportunity, against 2% for Nurture

57%

of all qualified leads sit in the top 20% of scores, beating the 50% target

0.97

ranking quality (AUC) on a later time period the model never saw

Results come from synthetic data with patterns we designed, so they validate the method rather than predict real-world performance.

Sales treats every lead the same, so the best ones wait

29%

of leads with a recorded outcome were spam

35%

of all leads never got an outcome recorded

25 h

average wait for a first reply, for good and bad leads alike

3%

of leads that later qualified were answered within 2 hours

Problem statement

The company receives more inbound leads than sales can review carefully. Spam, vendor pitches and unqualified inquiries sit in the same queue as teams ready to launch a card program or lending product.

With no consistent way to tell them apart early, high-intent leads wait as long as junk, and the team has no shared definition of a good lead.

How might we

make sure the next lead a salesperson opens is the one most likely to become a customer, and show them why?

The cost of the status quo shows up in three places

25 h

Time spent on junk

At 10 minutes per spam lead, the 150 tagged spam leads alone cost about 25 sales hours a year. Untagged spam adds more that nobody counted.

24.5 h

Slow response to buyers

Leads that went on to qualify waited 24.5 hours on average, the same as every other lead. In infrastructure sales, the first credible reply often wins the evaluation.

276

Lessons never learned

276 leads have no recorded outcome. Without outcomes, nobody can tell which channels, products or messages bring real buyers, and marketing budget follows guesses.

Hours per spam lead is an assumption to validate with sales. Response times and counts come from the Phase 0 audit of the synthetic dataset.

The first move was to split it into two different problems

Spam gate

Is this a real person with a real need?

Type	Yes or no filter
Worst mistake	Blocking a real buyer. One lost deal outweighs hundreds of spam leads that slip through.
Design choice	Quarantine, never delete. Weekly human review. Precision before recall.

Conversion score

Among real leads, who deserves time first?

Type	Ranking plus a calibrated probability
Worst mistake	A hot lead left waiting while sales works a weak one.
Design choice	Buckets with response targets, plain-language reasons, and a way to say the score is wrong.

Treating these as one model would force a single threshold to do two jobs with opposite error costs. Splitting them lets each stage be tuned for the mistake it must avoid.

We defined success in sales outcomes, not model accuracy

What we predict

Spam	Bots, fake identities, scams, vendors pitching their own services
Unqualified	Real person, no buying need
Qualified+	Reached a qualified opportunity. This is the primary label.
Converted	Signed contract. The long-term label.

Infrastructure deals take months to close. Training on signed contracts alone would give the model few, very late examples, so we train on Qualified+ and check each quarter that it tracks revenue.

How we judge it

Metric	Target	Prototype
Qualified leads in top 20% of scores	50% or more	57%
Genuine leads wrongly quarantined	Under 2%	0%
Hot leads answered within 2 hours	Most of them	3% today
Sales hours on spam	Down 70%	Needs live data
Sales override rate	Falling over time	Tracked in prototype

We shipped in phases, each useful on its own

Phase0

Labels and logging

Agree definitions, make outcome tagging mandatory, measure baselines.

Exit when 90% of new leads tagged

Prototyped

Phase1

Spam gate

Rules plus an LLM classifier, quarantine queue, weekly review.

Exit when under 2% good leads blocked

Prototyped

Phase2

Points score

Transparent weights agreed with sales, buckets and reasons.

Exit when top bucket clearly beats average

Prototyped

Phase3

Learned model

Logistic regression on outcomes, calibrated probability, retraining.

Exit when beats phase 2 on unseen data

Prototyped

Product

Sales experience

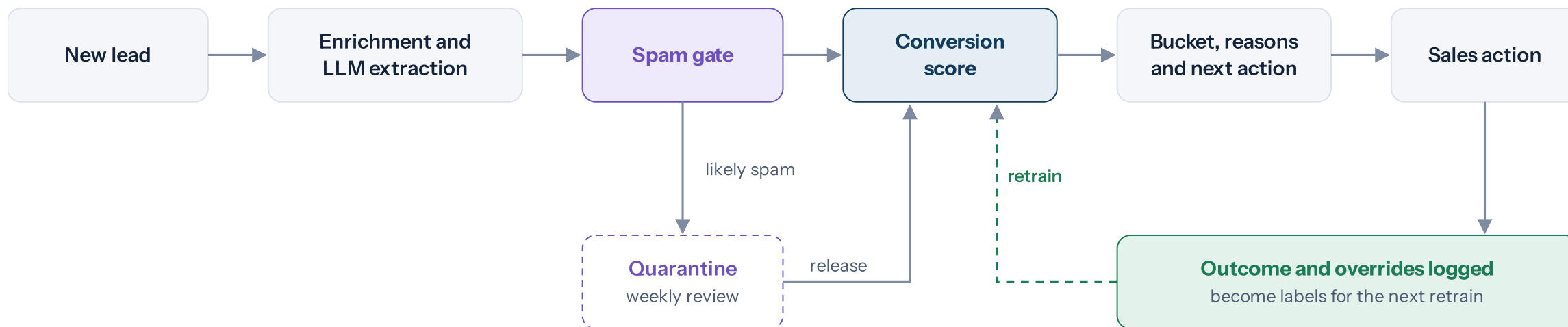
Queue, quarantine review, feedback loop, model health view.

Exit when sales uses it daily

Prototyped

Phase 2 exists on purpose. A points score sales helped design earns trust early and produces cleaner labels than fitting a model to messy history on day one.

One pipeline, with a feedback loop at its core



LLMs read, models predict

The LLM turns free-text form messages and company websites into structured signals like stated use case or vendor pitch. The probability itself comes from a model trained on the company's own outcomes, because direct LLM estimates are not calibrated.

Only arrival-time signals

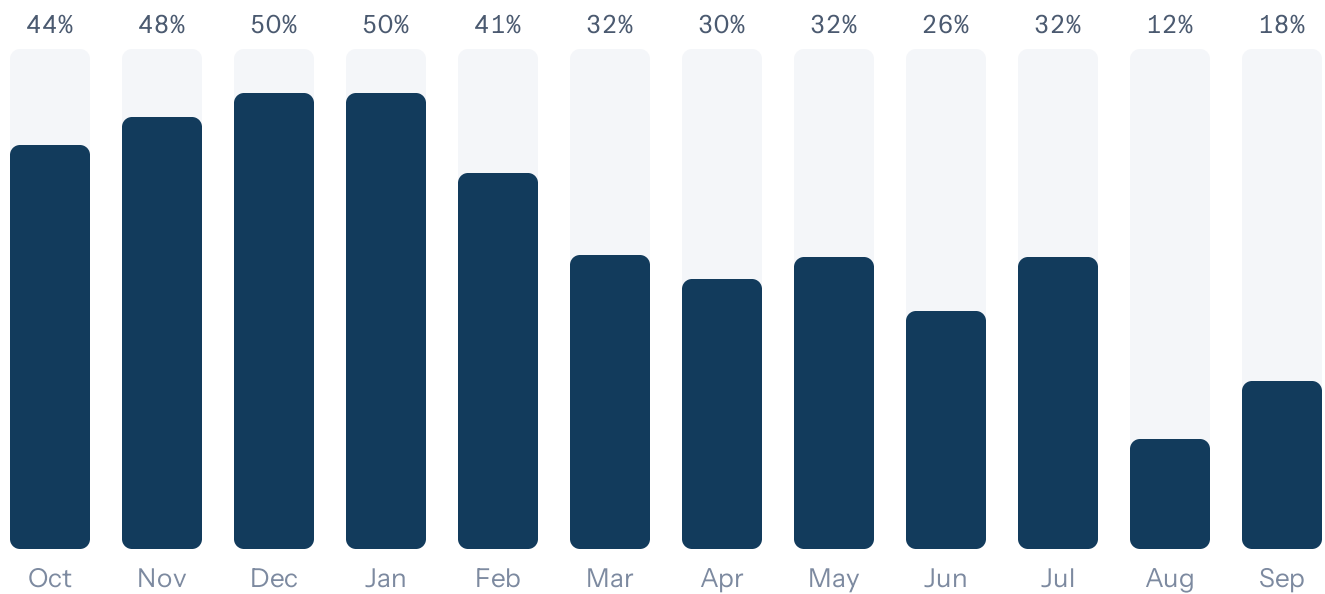
Every feature must be known when the lead arrives. Signals like a booked meeting would leak the answer into the prediction and inflate test results.

Humans stay in the loop

Released quarantine leads are logged as rule misses. Flagged scores are logged with a reason. Both become training data, so the system improves from normal sales work.

Phase 0 showed the data problem comes before the model problem

Share of leads never tagged with an outcome, by month



Oct 2025 to Sep 2026. Older leads are least likely to have an outcome.

What we did about it

Made outcome tagging mandatory when a salesperson closes out a lead, with a short fixed list of statuses.

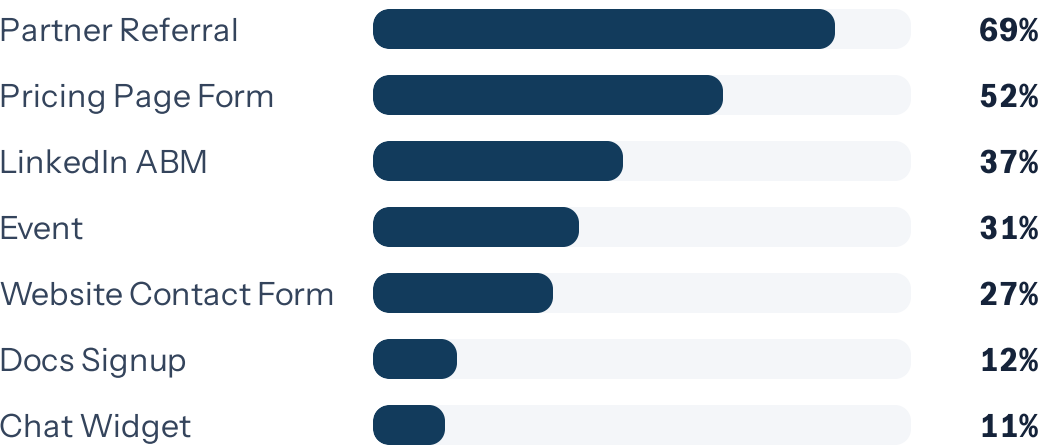
Hand-labeled a 150-lead sample to create a clean evaluation set for the spam gate.

Measured baselines for every success metric before building anything, so later gains can be proven.

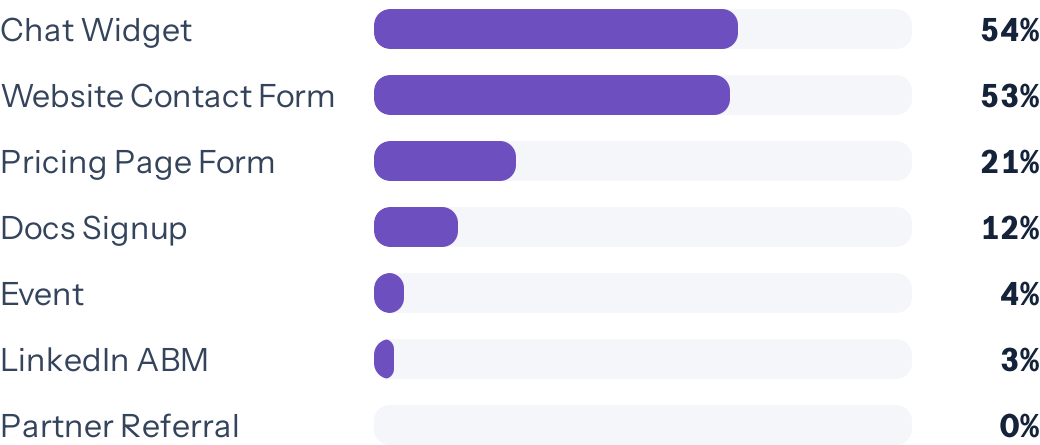
A model trained on this history would learn the logging habits of the team as much as the behavior of buyers.

Where a lead comes from predicts most of its value

Qualification rate of resolved genuine leads



Spam share of tagged leads



60% vs 10%

qualification rate for company email versus free email

79 to 95%

qualification rate for lending and virtual accounts

17%

qualification rate for KYC and data API interest

Precision first, so two plausible spam rules had to go

Rule	Precision	Recall	Decision
Disposable email domain	100%	28%	Kept
Vendor or scam wording	100%	61%	Kept
Very short message	100%	39%	Kept
Repeat submission	81%	37%	Removed
Free email and no company	27%	17%	Removed

Precision is the share of flagged leads that were really spam. Recall is the share of all spam the rule catches. Measured on leads with a recorded outcome.

34%

of flagged leads were genuine with all five rules

0%

after removing the two weak rules, with every tagged spam lead still caught

Both removed rules sounded reasonable. Plenty of real founders use Gmail, and real buyers do submit twice. A rule earns its place through measured precision, not intuition.

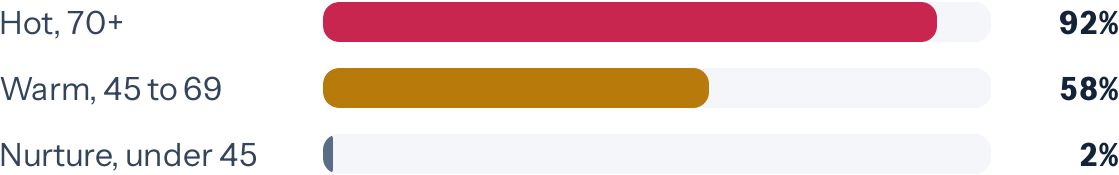
A transparent points score already separates leads sharply

Sample of the scoring weights

Partner Referral	+30
Pricing Page Form	+25
LinkedIn ABM	+20
Company email	+20
Payments or lending interest	+15
8+ pages visited	+15
Buying intent in message	+15
Negative intent in message	-30

Every weight is visible and editable. Each lead shows which signals fired, in plain words.

Qualification rate by bucket



70%

of all qualified leads land in Hot

3%

of qualified leads fall to Nurture

A learned model edges out the points score and adds real probabilities

Tested on Jul to Sep 2026, a period the model never saw

Measure	Model	Points
Ranking quality (AUC)	0.972	0.966
Qualified leads in top 20%	63%	59%
Probability error (Brier)	0.081	n/a
Brier if everyone got the average	0.220	n/a

Logistic regression trained on 245 resolved leads, tested on 83. The ranking gap is within noise at this size. The real gain is a probability that means something.

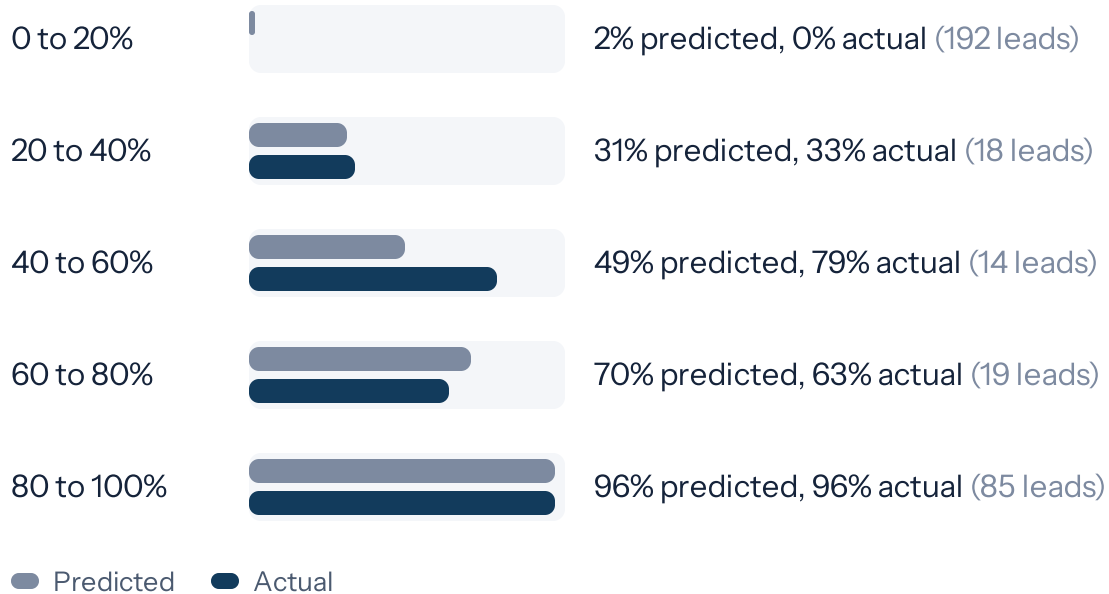
What the model learned, in log-odds



Green pushes the probability up, red pushes it down. Weights came from outcomes, not from a meeting.

When the model says 96%, about 96% of those leads qualify

Predicted probability versus actual qualification rate



Why calibration matters to sales

A score of 73 means nothing to a salesperson. A probability of 73% that behaves like one lets them plan their day, and lets a manager forecast pipeline from the queue.

The two largest bands, holding most leads, line up almost exactly. The middle bands hold only 14 to 19 leads each, so their gaps are mostly noise.

Watch the 40 to 60% band in shadow mode. If real data shows the same under-prediction, it is a signal the model is missing a feature.

Sales sees a queue with reasons, not a bare number

Lead Queue

Prototype, dummy data

Queue

27

Quarantine

9

Model health

Who to contact next

Call these first. Target reply within 2 hours.

Hot 5

Warm 12

Nurture 10

Ananya Nguyen at Hollowquay Bank

Pricing Page Form, Payments - Cross-Border

Strong source. Company email. High-value product. High engagement. Buying intent in message.

Tariq Khan at Rushbarrow Capital

Website Contact Form, Payments - Cross-Border

Company email. High-value product. High engagement. Buying intent in message.

Fatima Khan at Ostrivo Studio

LinkedIn ABM, Payments - Payouts

Strong source. Company email. High-value product. Buying intent in message.

Lina Novak at Marrowgate Pay

Pricing Page Form, Payments - Cross-Border

Strong source. Company email. High-value product. Buying intent in message.

Zara Nguyen at Ferrowind Capital

LinkedIn ABM, Payments - Virtual Accounts

Strong source. Company email. High-value product. Buying intent in message.

Arrival times are compressed into the last three days for the demo. Actions are kept for this session only.

Hot

Close

Ananya Nguyen

Hollowquay Bank, Vietnam

>99% likely to qualify

Almost every similar lead reached a qualified opportunity.

Migrating from another provider for our card program, need a quote this month.

Why this score

Source

+25

Email

+20

Product

+15

Engagement

+15

Buying intent

+15

Points total

90

Details

Source

Pricing Page Form

Interest

Payments - Cross-Border

Email

ananya.nguyen@hollowquaybank.example

Pages seen

11

Waiting

26 h

Lead ID

L-0754

Mark qualified

Mark unqualified

Flag this score

- 1

Buckets with response targets. Hot means reply within 2 hours.
- 2

Reasons in plain words on every lead, so the score can be trusted or challenged.
- 3

Probability read as odds, plus the points that produced it.
- 4

One click to record an outcome or flag a wrong score. This is the feedback loop.

Quarantine and model health keep the system honest

Quarantine

Leads the spam rules held back. Nothing here is deleted. Release anything real and it goes to Warm for a human look, and the release is logged as a rule miss.

Ananya Kim

ananya268@burnmail.example, Website Contact Form

"hi"

Disposable email

Very short message

Release to queue

Confirm s

Noah Sharma at Rankfarrow SEO Agency

noah801@webmail.example, Docs Signup

"Exclusive investment opportunity with 300% returns, DM on telegram."

Vendor or scam wording

Release to queue

Confirm s

Ravi Berg at Hypemarrow PR

ravi714@hypemarrowpr.example, Website Contact Form

"Boost your Twitter followers and Telegram members, organic fintech marketing."

Vendor or scam wording

Release to queue

Confirm s

Ivan Haddad at Codewhistle Outsourcing

ivan363@codewhistleoutsourcing.example, Website Contact Form

"Boost your Twitter followers and Telegram members, organic fintech marketing."

Vendor or scam wording

Release to queue

Confirm s

Quarantine review

Every held lead shows the rule that fired. Releasing a lead sends it to Warm and logs a rule miss, which is how the spam gate gets tuned.

Model health

Is the score still telling the truth? Based on all leads with a recorded outcome.

Qualification rate by bucket

Hot		92%
Warm		58%
Nurture		2%

Hot holds 70% of all qualified leads. The top 20% of scores hold 57%, against a target of 50%.

Does the probability match reality?

0% to 20%		2% vs <1% (192)
20% to 40%		31% vs 33% (18)
40% to 60%		49% vs 79% (14)
60% to 80%		70% vs 63% (19)
80% to 100%		96% vs 96% (85)

■ Predicted ■ Actual

Bars close together mean the percentage can be trusted. Small bands are noisy.

Baseline before s

25 h

average first reply

29%

of tagged leads were spam

Feedback this session

No feedback yet. Mark outcomes or flag a score in the queue and it shows up here.

Every outcome and flag becomes a training label at the next retrain.

Model health

Bucket hit rates, calibration and the pre-launch baseline in one view, so drift is visible before sales stops trusting the score.

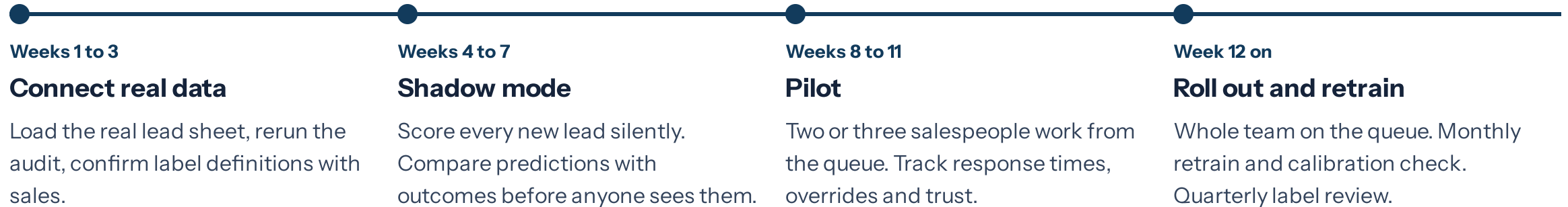
Product

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The biggest risk is a system that shapes the outcomes it predicts

Risk	What goes wrong	Mitigation
Self-fulfilling loop	Only Hot leads get worked, so only Hot leads convert and the model looks right	Hold out a small random sample of low-scored leads for full follow-up
Good leads marked as spam	Lost revenue that nobody sees	Quarantine not delete, weekly review, precision-first rules
Too few outcomes	Unstable weights and noisy tests	Start with points, use Qualified+ as the label, keep models simple
Long sales cycles	The model learns slowly	Proxy label now, quarterly check against signed revenue
Stale enrichment	Wrong company fit signals	Freshness requirement, re-enrich whenever a lead is re-scored
Low sales trust	Scores ignored, feedback stops	Visible reasons, co-designed weights, a one-click override

Next, prove it on real leads before sales depends on it



Decision gates

Move to pilot only if shadow mode keeps wrongly quarantined leads under 2% and the top 20% of scores holds at least half of qualified leads. Move to rollout only if pilot reps answer Hot leads faster and override fewer than one in five scores.

Asks

Thirty minutes with sales to sign off definitions and weights. Access to the lead sheet or a monthly export. An owner for the weekly quarantine review.

Method, caveats and what was delivered

Method and caveats

All results use 800 synthetic leads from Oct 2025 to Sep 2026. We designed the patterns in that data, so strong results prove the pipeline works, not that real leads behave this way.

Phase 2 weights were drafted on the same data they were tested on, so its numbers are optimistic. Phase 3 was trained before Jul 2026 and tested after, but the test set holds only 83 resolved leads.

Response times and the hours spent per spam lead are baseline estimates to replace with measured values.

What was delivered

PRD	Goals, definitions, requirements, evaluation plan and risks
Scoring workbook	Dummy leads, Phase 0 audit, spam rules, points score and live model probabilities, all formula driven
train_model.py	Trains, tests on a later period, compares with the points score, exports weights
Queue prototype	Clickable sales experience with quarantine and model health views